

# Hamiltonian-Geometric Optimization Framework for Machine Learning

## A Concise Walkthrough: Framework, Validation, Evidence

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Capacitor Fringes Project

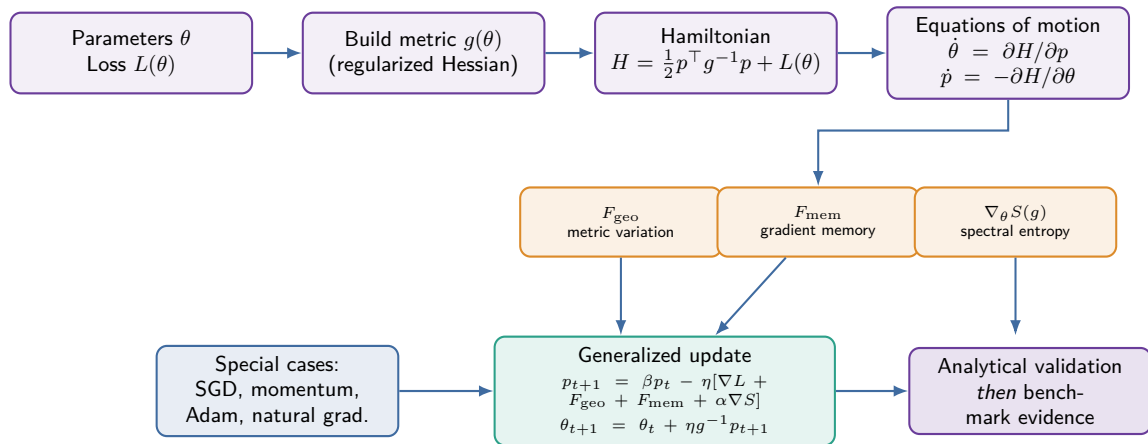
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## Goal

Can a single **Hamiltonian-geometric** update rule act as a useful optimizer family across classical ML, LLM-style training, PINNs, and quantum-control objectives?

- Treat parameters  $\theta$  as generalized coordinates, the loss  $L(\theta)$  as potential energy, and optimizer state as momentum  $p$  evolving on a **metric manifold**  $g(\theta)$ .
- One family should recover SGD, momentum, Adam, and natural gradient as special cases.
- Claim is deliberately *bounded*: not uniformly superior, strongest when full-metric curvature is exploitable.

# Methodology — Mathematical Framework at a Glance



Deterministic stages left to right; forces merge into one update rule; output feeds validation before benchmarking.

$$H(\theta, p) = \frac{1}{2} p_i g^{ij}(\theta) p_j + L(\theta),$$
$$\dot{\theta}^i = \frac{\partial H}{\partial p_i} = g^{ij}(\theta) p_j, \quad \dot{p}_i = -\frac{\partial H}{\partial \theta^i} = -\partial_i L - \frac{1}{2} p_j p_k \partial_i g^{jk}(\theta).$$

- Second term in  $\dot{p}_i$  is the **geometric force** from a coordinate-dependent metric.
- Dissipation is added so dynamics *optimize* rather than conserve energy.
- Same structure carries the memory and spectral-entropy correction terms.

# One Update Rule, Many Optimizers

| Optimizer             | Hamiltonian-geometric specialization                                     |
|-----------------------|--------------------------------------------------------------------------|
| SGD / heavy-ball      | Flat metric, $F_{\text{geo}} = 0$ , instant or finite momentum relax     |
| Adam / RMSProp        | Diagonal adaptive metric; ignores $F_{\text{geo}}$ from metric variation |
| Natural gradient      | Full information metric, usually no momentum or memory force             |
| Hamiltonian-geometric | Full metric <i>with</i> momentum, memory, spectral entropy, curvature    |

- Modern optimizers (Shampoo, K-FAC, Muon, AdaFactor, LARS/LAMB) sit at intermediate points on the same metric-structure hierarchy — from full  $n \times n$  down to a single global scalar.

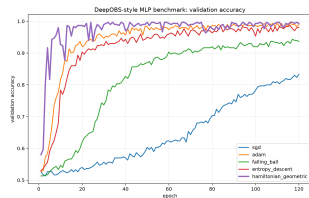
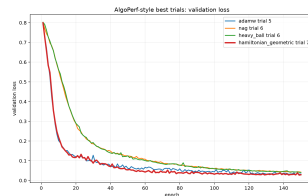
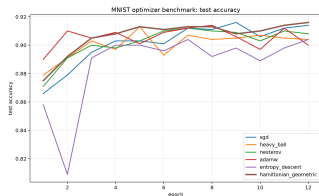
# Analytical Validation (Equation-Level, All PASS)

**What & reference:** each row checks one equation from Sparsho Chakraborty's *literature-review PDF* — the design reference for this optimizer's architecture — against a finite-difference computation on a fixed, seeded, genuinely curved-metric toy problem.

| Check                                | Kind       | Max error              | Tol.      | Result |
|--------------------------------------|------------|------------------------|-----------|--------|
| Legendre transform                   | literal    | $2.96 \times 10^{-11}$ | $10^{-6}$ | PASS   |
| Geometric force                      | literal    | $4.60 \times 10^{-11}$ | $10^{-5}$ | PASS   |
| Flat-metric reduction                | literal    | 0.000                  | $10^{-9}$ | PASS   |
| Memory recursion                     | literal    | $8.88 \times 10^{-16}$ | $10^{-9}$ | PASS   |
| Hessian regularization               | counterex. | 0.000                  | $10^{-9}$ | PASS   |
| Rayleigh dissipation (curved metric) | completed  | $1.64 \times 10^{-9}$  | $10^{-6}$ | PASS   |
| Spectral entropy gradient            | completed  | $1.14 \times 10^{-10}$ | $10^{-5}$ | PASS   |

- **literal** = tests a PDF equation exactly as printed; **completed** = tests our documented completion of a formula the PDF states but never fully derives; **counterex.** = shows the PDF's literal condition fails, then confirms the shipped code fix succeeds.

# Benchmark Evidence Snapshot



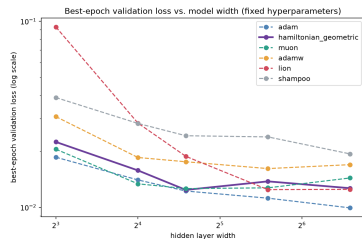
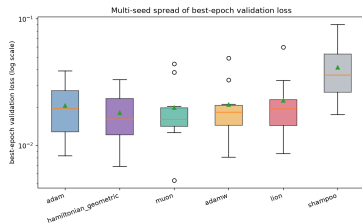
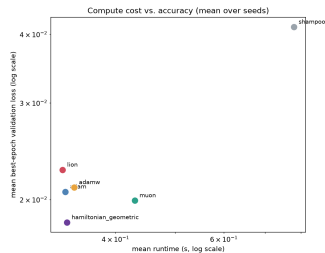
## Benchmark Results — Classical, PINN & LLM

| Workload                         | Hamiltonian-geometric    | Best other optimizer              | Reading                      |
|----------------------------------|--------------------------|-----------------------------------|------------------------------|
| MNIST softmax (test)             | Loss 0.2938, acc. 0.9160 | Nesterov: loss 0.2962             | Best                         |
| AlgoPerf-style MLP (val.)        | Loss 0.0347, acc. 0.9867 | AdamW: loss 0.0430                | Best                         |
| DeepOBS-style MLP (val.)         | Loss 0.0203, acc. 0.9933 | Adam: loss 0.0449                 | Best                         |
| Capacitor PINN (final)           | Loss 17.13218            | Entropy descent: 17.31064         | Best                         |
| WikiText-2 LLM (val., 200 steps) | Loss 6.778, ppl. 878.3   | AdaFactor: loss 6.765, ppl. 866.8 | 2nd best, fastest wall-clock |

- On the LLM run, AdaFactor edges it out on loss, but Hamiltonian-geometric clearly beats AdamW (6.843) and Lion (7.032), and finishes fastest of the five; Muon-lite diverges (loss 9.31) under this untuned configuration.
- Small-scale, single-seed runs throughout — evidence of mechanism, not a scaling proof.



# Modern Optimizer Comparison



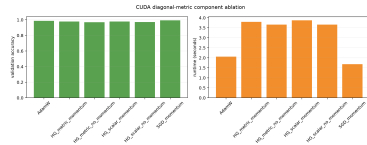
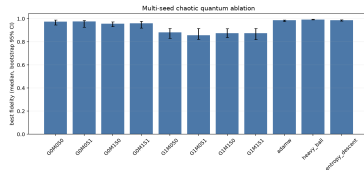
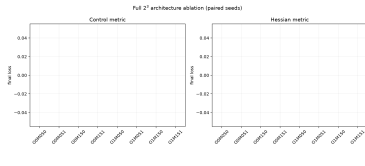
Adam, Hamiltonian-geometric (diagonal-metric reduction), and Muon lead the pack; the 10-seed sweep (center) holds the near-tie, but the width-scaling sweep (right) favors Adam — competitive evidence, not decisive dominance.

## Benchmark Results — Modern Optimizers (Spiral MLP)

| Optimizer                    | Best epoch | Val. loss | Val. acc. |
|------------------------------|------------|-----------|-----------|
| Adam                         | 102        | 0.012244  | 0.9967    |
| <b>Hamiltonian-geometric</b> | 120        | 0.012416  | 0.9967    |
| Muon                         | 102        | 0.012624  | 0.9967    |
| AdamW                        | 91         | 0.017588  | 0.9967    |
| Lion                         | 113        | 0.018797  | 0.9967    |
| Shampoo                      | 117        | 0.024266  | 0.9933    |

- Ranked by best-epoch validation loss, lower is better.
- Adam, Hamiltonian-geometric, and Muon differ by at most  $4 \times 10^{-4}$  in absolute validation loss — a narrow leading group, not a clearly separated ordering.
- AdamW and Lion trail by a larger margin; Shampoo is both slower and slightly worse here.

## Extended Ablation Study — Design & Evidence



A separate, more rigorous  $2^3$  factorial (geometric  $\times$  memory  $\times$  spectral force), paired-seed replication: 40 seeds/300 steps (architecture, left), 10 seeds/40 iterations (chaotic quantum, center), 10 seeds/600 steps (CUDA diagonal reduction, right).

# Extended Ablation Study — Key Numbers

| Study                               | Factor                  | Effect      | $p$                   |
|-------------------------------------|-------------------------|-------------|-----------------------|
| Architecture, control metric        | $G, M, S$               | $\leq 0.59$ | all $> 0.2$ (n.s.)    |
| Architecture, <b>Hessian metric</b> | $S$ ( <b>spectral</b> ) | 2.08        | $9.25 \times 10^{-5}$ |
| Architecture, Hessian metric        | $G, M$                  | $\leq 0.32$ | $> 0.5$ (n.s.)        |
| Quantum (kicked-top)                | $G$ (geometric)         | 0.88        | $1.57 \times 10^{-4}$ |

- **Architecture (40 seeds):** confirms the earlier story — the spectral term only matters when the metric is genuinely tied to loss curvature (Hessian), not under an unrelated control metric.
- **Quantum (10 seeds, paired):** here the honest result is negative — turning on the geometric force *significantly lowers* fidelity (0.86–0.88 vs. 0.96–0.98 with it off), and every simple baseline (heavy-ball 0.993, entropy descent 0.988, AdamW 0.986) beats every full-HG variant, while costing 10–20 $\times$  more runtime.
- **CUDA scaling (10 seeds):** the diagonal-metric Torch reduction (0.978 acc.) trails plain SGD (0.993) and AdamW (0.987) here too, at roughly 2 $\times$  the runtime.
- **Rigor rewards honesty:** this replication does not uniformly favor Hamiltonian-geometric — it sharpens exactly where the framework helps (curvature-tied spectral term) and where it currently does not (quantum control, GPU-scale diagonal reduction).

# Limitations

- **Scale:** all systems are small; full  $n \times n$  metric is mechanism evidence, not a production-scale replacement for Adam-family optimizers.
- **Statistics:** the earlier classical/LLM benchmarks are mostly single-seed; multi-seed variance now comes from the modern-optimizer sweep (10 seeds) *and* the extended ablation (40/10/10 seeds) — and the latter is where the negative quantum and GPU-scaling results appear.
- **Tuning:** learning rates from modest grids, not exhaustive search.
- **Comparability:** the modern-optimizer and CUDA-ablation runs both use a diagonal reduction, not the full curvature-memory-entropy architecture.

- Formulation is mathematically coherent once the spectral-force sign convention and memory-metric construction are made explicit.
- Evidence is strongest where exploitable full-metric geometry is present.
- Places the method in the leading group of modern optimizers, without separating it decisively from Adam or Muon.
- The rigorous extended ablation confirms this precisely: the spectral term earns its keep only under a genuine Hessian metric, while quantum and GPU-scale results caution against over-claiming.

**Thank You**

Hamiltonian-Geometric Optimization Framework